

Unequal Job Opportunities and Well-Being Inequality

A Latent-Jobs Structural Decomposition

Hisham Haydar

University of Luxembourg & LISER

4th-Year PhD Workshop

Discussant: Sebastian Dobre

Thank you, and thank you to Sebastian for discussing the paper. This talk asks how much of the inequality in money-metric well-being is associated with unequal job opportunities rather than with differences in what people prefer. I build the answer in four steps: a model of labour supply as choice among latent jobs, which separates preferences from opportunities; two ways of turning choices into euros of well-being; counterfactuals that equalise one source of heterogeneity at a time; and a Shapley rule that allocates the change. The punchline is that the answer depends on the welfare question: prospects or attained outcomes.

Observed		Well-being
C_i		M_i
Gini of consumption		Gini of the attained-bundle money metric
single adults 0.2633 couples 0.2268		single adults 0.2498 couples 0.1974
low earnings ⇐ tastes or opportunities or resources		
equivalised, household-weighted, within each population		

Two people can work the same hours for the same hourly pay and be very differently placed. One chose that job from several that were open; the other took the only offer available. Their incomes are identical and their circumstances are not.

Low earnings can describe someone who values leisure and chose short hours, someone who cannot obtain a well-paid job, or someone whose household resources make a different arrangement affordable. An income distribution records outcomes with these explanations already mixed in.

Once the leisure a job costs is valued, the dispersion of well-being is not the dispersion of consumption. These are within-population comparisons, and single-adult and couple households are separate populations that I never compare with each other.

The conflict

Compensation	Responsibility
unequal job opportunities should not become unequal well-being	preference-based choices should not be compensated

Compensation vs. Responsibility

$\nexists W : \text{Full Compensation} \wedge \text{Full Responsibility}$

[Fleurbaey & Maniquet]

Before any data, the conflict that makes this paper necessary.

A fair well-being comparison would like to respect two principles. The first is compensation: people should not end up worse off because they face worse job opportunities. The second is responsibility: people should be held responsible for their preferences, so a choice that reflects taste should not generate a compensation claim.

Each principle is attractive on its own, but they cannot both be imposed in full: no well-being measure satisfies full compensation and full responsibility together. The literature, following Fleurbaey and Maniquet, is organised around compromises.

To know what to compensate, we must separate what people prefer from what they can reach. And observed choices mix the two.

How much inequality in money-metric well-being is associated with unequal job opportunities rather than heterogeneous preferences, once labour supply is modelled as choice among latent jobs?

1. Do observed labour-supply choices reflect preferences alone, or also heterogeneous latent job opportunities?
2. How do conclusions differ when welfare evaluates the attained bundle versus the ex-ante opportunity prospect?
3. In a restricted P/A/B decomposition, how much welfare inequality is associated with preferences, local access, and earning opportunities?

The whole talk on one slide. The first subquestion is behavioural: can we read preferences off choices, or do opportunities get in the way? The second is normative: once well-being is valued in euros, does it matter whether we value the prospect a household faces or the outcome it attains? The third is the accounting: in a restricted decomposition with preferences, local access and earning opportunities as the three channels, how is inequality allocated?

Note the word associated. Nothing in this talk is a causal claim.

Latent jobs	Money metrics	Decomposition
Aaberge, Dagsvik & Strøm	Fleurbaey & Maniquet	Shorrocks
Dagsvik & Jia	Decoster & Haan	Creedy & Hérault
Capéau, Decoster & Dekkers	Jacquet, Jia & Thoresen	Mühlhan

reforms and *changes* $\longrightarrow$ a *level* of well-being inequality

preferences vs. opportunities, inside an estimated latent-jobs model

Every ingredient has antecedents; the contribution is the combination and the empirical answer.

Labour supply as choice among latent jobs is due to Aaberge, Dagsvik and Strøm, developed by Dagsvik and Strøm and by Dagsvik and Jia, and set out for applied work by Capéau, Decoster and Dekkers, who already cover singles and couples. Money-metric evaluation with such models exists, but it evaluates a change between policy regimes. The normative literature following Fleurbaey and Maniquet shows how the reference encodes compensation and responsibility. The closest decompositions, Creedy and Hérault and Mühlhan, decompose a change between situations. Our object is a cross-sectional level of well-being inequality. The Shapley rule is inherited from Shorrocks.

$j = (e, k, h, w)$ 4 occupation groups $C_i(j)$ priced through EUROMOD

$$v_i(j) = L_i(j) + \beta_c \log\left(\frac{C_i(j)}{\lambda_c}\right)$$

L_i
value of leisure: age; children for
women

β_c
weight on log consumption,
estimated

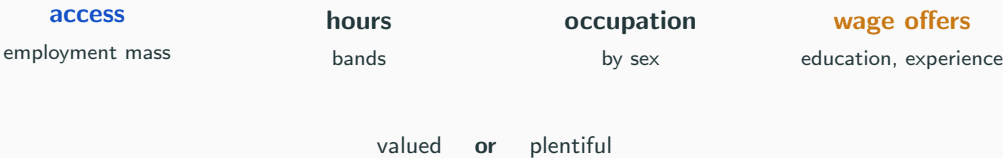
λ_c
units convention

In words: a household's systematic value of a job is the value of the leisure it leaves plus the log of the consumption it pays.

A job is a package: an employment state, an occupation group, weekly hours and an hourly wage. Every package is run through the French tax-benefit system, so consumption is the household's actual disposable income at that job. A couple chooses a pair of packages under a shared budget, and the leisure term sums both spouses' terms. The household-specific leisure coefficients are the preference channel; the child term is a reduced-form time-constraint shifter, not pure taste.

Opportunities: the choice probability

$$P_i(j) = \frac{\exp\{v_i(j)\} g_i(j)}{\int \exp\{v_i(k)\} g_i(k) d\nu(k)}$$



In words: the probability of being seen in a job is its value times its availability, normalised over all jobs.

The opportunity density g says how intensely a package is available before any choice. A job is chosen often when it is valued or when it is plentiful. If everyone faced the same density, every difference in behaviour would have to be a difference in taste. Because the density is household-specific, part of behaviour can be attributed to reachability.

Access moves with local unemployment exposure, region, urban or rural location, and year. Hours are a step function over bands, occupation availability is sex-specific, and wage offers are log-normal, located by education and experience. Separation rests on maintained restrictions, not tested ones.

EU-SILC France

collected 2016, incomes 2015

1,540 single-adult households

2,223 couple households

EUROMOD

French 2015 policy system

$h \in [5, 70]$ $w \in [2, 590]$

100 priced alternatives per household

The data are French EU-SILC, with incomes referring to the year before collection. Taxes, benefits and disposable income come from EUROMOD under the policy system in force over the income period.

The samples are single-adult and couple households of working age, not in education and not receiving a pension; the two samples are estimated separately. Hours are in hours per week and wages in euros per hour. Each household’s likelihood is evaluated on its observed job plus sampled alternatives, each run through the tax-benefit model. Sampling is a computational device, not an offer.

Household-specific opportunities

free coordinates 41

population fit 0.0140

Common opportunities

free coordinates 10 / 16

population fit 0.0273 / 0.0264

criterion worse by at least 141.64

choices reflect opportunities, not preferences alone

single adults; population fit is the mean absolute deviation of predicted from observed moments

The first subquestion. Take away the household-specific opportunity block, re-estimate everything else, and the model does worse on the same households, the same sampled alternatives and the same criterion.

The deterioration is concentrated where the opportunity block does its work: population fit roughly doubles its error. I do not turn this into a likelihood-ratio test, because the sampled-alternative criterion is not the likelihood of the observed data. Fit is not perfect: the observed concentration at exactly thirty-seven hours is underpredicted in every group, and extensive-margin accuracy for men is withheld because integration error is too large. The buttons open the parameter tables, the fit evidence and the identification argument.

Selected estimates

	Single adults	Couples
β_c consumption weight	2.0387 (0.2917)	2.1017 (0.2939)
$\beta_{E,s}$ local unemployment exposure, in access	−1.4422 (0.2356)	−1.1924 (0.1557)
β_{wH} high education, wage-offer location	0.1491 (0.0308)	0.1817 (0.0197)
σ log-wage dispersion	0.3815 (0.0133)	0.3631 (0.0072)

cluster-robust standard errors in parentheses; the two populations are estimated separately

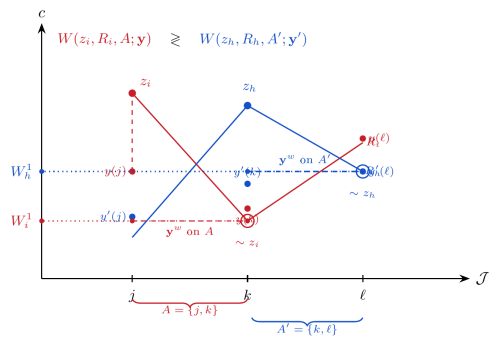
Four estimates that carry the economics; the full vectors are in the backup.

The consumption weight is positive and similar across the two separately estimated models. It is the rate at which a household's non-consumption advantage converts into a proportional consumption adjustment in both welfare measures.

Higher local unemployment exposure shifts employment opportunity mass downward, for single adults and for couples.

Higher education shifts the wage-offer distribution upward, and log-wage dispersion is common within each model.

These are structural associations inside the estimated model, not causal estimates.



Own-set equal-consumption equivalents: each individual gets a common consumption on every job in their own ability set; the level at which the preferred job is indifferent to the attained bundle is their money metric, comparable across individuals. Adapted from Haydar and Maniquet (2026), work in progress.

Estimated utility is not an interpersonal welfare index: preferences differ, so the same utility number means different things for different people. Each situation is translated into euros through an indifference condition with an explicit reference, and the choice of reference is the welfare question.

Both money metrics descend from this construction in the companion theory paper. Each individual is evaluated with their own preferences and their own set of jobs. Give every job in that set the same consumption; the consumption at which the preferred job is exactly as good as the attained bundle is their money metric. The next two slides take the construction to the data in two ways.

$$J_i = \int e^{L_i(j)} \left(\frac{C_i(j)}{\lambda_c} \right)^{\beta_c} \widehat{g}_i(j) d\nu(j), \quad H_i = \int e^{L_i(j)} \widehat{g}_i(j) d\nu(j),$$

$$M_i^{\text{EA}} = \lambda_c \exp \left\{ \frac{\log J_i - \log H_i}{\beta_c} \right\}$$

J_i

the actual prospect

H_i

same jobs, equal pay

M_i^{EA}

flat-consumption equivalent

In words: EA is the constant monthly consumption at which a prospect with the household's own jobs and availability, but equal pay everywhere, is worth exactly as much as the prospect it actually faces.

Every job enters J weighted by how available it is and how much it is valued; H keeps the same jobs, availability and preferences with pay differences removed. Opportunities enter welfare directly: access moves which jobs carry weight; earning opportunities move what they pay.

I show this perspective first because this talk is about opportunities; that is an order of presentation, not a ranking. The measure treats taste-shock variety as part of what makes a rich prospect valuable, which is a normative position.

$$M_i^{\text{att}} = C_i^{\text{obs}} \exp \left\{ \frac{L_i(j_i^{\text{obs}}) - L_i(o)}{\beta_c} \right\}$$

o

non-employment, reachable by all

j_i^{obs}

the only route for opportunities

In words: ATT is observed consumption, adjusted by the leisure the job costs relative to not working, converted at the rate one over the consumption weight.

The reference is non-employment, which every household can reach. On the current empirical domain the equal-consumption reference job is non-employment for every household, so the measure coincides with the staying-home equivalent; that is a property of this specification, not a theorem. Opportunities matter only through which job is attained and what it pays.

The two measures answer different welfare questions: one values the prospect, the other the attained outcome. Neither corrects the other.

Inequality and Shapley decomposition

P	A	B	held fixed
preferences: age, children	local unemployment exposure, region, urban or rural location, year	wage-offer location	household resources, needs and composition

$$\phi_k^P = \sum_{S \subseteq \{P,A,B\} \setminus \{k\}} \frac{|S|! (3 - |S| - 1)!}{3!} \left[v^P(S \cup \{k\}) - v^P(S) \right]$$

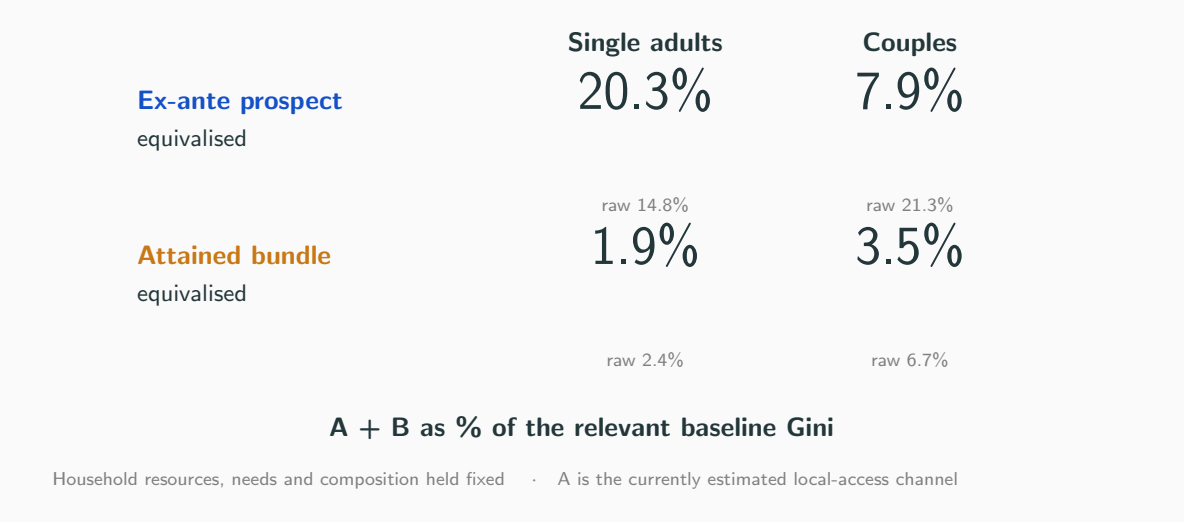
$$v^P(S) = I^P(\emptyset) - I^P(S) \quad I^P: \text{household-weighted Gini of } M^P$$

In words: a channel's Shapley contribution is its marginal effect on inequality, averaged over every order in which the three equalisations can be introduced; the three contributions add up exactly to the total change.

P equalises age profiles and the child-related shifter. A equalises local access: local unemployment exposure, region, urban or rural location, and year; it is local access, not total opportunity. B equalises the systematic wage-offer location, not wage dispersion. Household resources, needs and composition are held fixed in every coalition.

Each coalition is a full structural counterfactual: coefficients never change, welfare is recomputed for every household under each perspective, and inequality is measured again. Contributions are signed.

How much is associated with opportunities?



This is the first result. Each number is local access plus earning opportunities, the A plus B part of the restricted game, as a percentage of that perspective’s own baseline Gini, on the equivalised scale.

When welfare values the whole prospect, the two opportunity channels are associated with 20.3 per cent of baseline inequality for single adults and 7.9 per cent for couples. When welfare values the attained bundle they are associated with 1.9 and 3.5 per cent.

The small grey numbers are the raw household scale, for reference only. Household resources, needs and composition are held fixed, and A is the local access channel as currently estimated. Prospects versus attained outcomes: the two perspectives answer different welfare questions, and neither is designated primary. None of these is a causal share.

Which opportunity channel matters?



ex-ante prospect, equivalised: % of baseline Gini

single adults: access > earnings couples: earnings > access, close

attained bundle: earnings > access for both populations

Now split the ex-ante number into its two channels, each as a percentage of the baseline Gini on the equivalised scale.

For single adults, local access is the larger channel, about three times earning opportunities. For couples, earning opportunities are larger than access, but the two are close on this scale.

Under the attained-bundle perspective the ordering is the other way for single adults: earning opportunities are larger than access for both populations. The full tables, with Gini points and the raw scale, are one click away. These are accounting shares within a restricted game, not causal shares.

Where does baseline inequality go?

shares of baseline Gini · ex-ante prospect, equivalised · current restricted P/A/B game · A = current local-access channel

		Single adults	Couples
Preferences	ϕ_P/l_0	0.1%	−1.8%
Local access	ϕ_A/l_0	15.5%	3.7%
Earning opportunities	ϕ_B/l_0	4.8%	4.2%
Not allocated by this exercise	$1 - (\phi_P + \phi_A + \phi_B)/l_0$	79.6%	93.9%
Total		100.0%	100.0%

Couples: A + B alone = 7.9%; P + A + B = 6.1% because the preference contribution is negative (−1.8%).

Not allocated is not a channel: it holds household resources, needs and composition fixed, retains sex-specific preference and opportunity blocks, and excludes wage-draw luck and the common offer spread, which exceed the location differences equalised.

one decimal; the not-allocated row closes each column to the total

Here is the same ex-ante decomposition, laid out so that it visibly sums to the whole baseline Gini, on the equivalised scale.

Under the current restricted ex-ante decomposition, labour-market opportunities are associated with about 20.3 per cent of baseline welfare inequality for single adults and about 7.9 per cent for couples.

For couples the preference contribution is negative, so preferences plus access plus earnings is smaller than access plus earnings alone; the table shows the negative row rather than hiding it.

The last row is not a channel and it is not needs explaining the rest. It is what this exercise does not allocate: household resources, needs and composition are held fixed, sex-specific blocks are kept, and the wage-draw luck and common offer spread are not equalised; those are larger than the location differences that the access and earning operators remove. None of these shares is causal.

What is not claimed

Scope

Household resources, needs and composition
held fixed

Access: local access, not total opportunity

Preliminary decomposition

Not claimed

Not causal

No parameter uncertainty yet

Preferences are not responsibility

Neither perspective is designated primary

prospects versus attained outcomes: two different welfare questions

All the boundaries in one place.

The decomposition is preliminary: a restricted structural accounting exercise, and the attained-bundle counterfactual integration is still being validated, with short hours sparsely covered. Nothing is causal: separation rests on maintained functional-form and exclusion restrictions. Parameter uncertainty has not been propagated through either decomposition. The preference channel is not a responsibility channel; deciding what people should be held responsible for is a normative step this paper does not take. Household resources, needs and composition are held fixed, and access is local access, not total opportunity. The two welfare perspectives answer different questions, and choosing between them is a normative decision not made here.

1. Choices reflect opportunities, not preferences alone.
2. Prospects: access leads for single adults. Attained outcomes: earnings lead. Couples: no reversal.
3. Restricted P/A/B: a bounded part of well-being inequality; preference sign not robust.

next: resources and needs as a fourth channel · parameter uncertainty · validation

To close, the three answers.

First, observed choices carry information about opportunities: a model that gives everyone the same jobs fits worse on the same households.

Second, the importance assigned to different labour-market inequalities depends on whether welfare evaluates the opportunity prospect or the realised outcome. Access leads for single adults when we value the prospect; earning opportunities lead when we value the attained bundle; couples do not reverse.

Third, within the restricted decomposition, local access and earning opportunities are a bounded part of well-being inequality, and the preference contribution is not sign-robust to equivalisation.

Next steps: household resources, needs and composition as a fourth channel; propagating parameter uncertainty; completing validation of the attained-bundle integration. Neither perspective is designated primary. Thank you.

Appendix map

Model	Utility specification	Opportunity density	Likelihood	Identification
Estimation	Preferences	Employment access	Hours and occupation	Wage offers
	Common opportunities	Inference		
Fit	Extensive margin	Fit by margin	Thirty-seven-hour point	Matched households
	Marginal rate of substitution			
Welfare	ATT derivation	EA derivation	EA validation	Equivalisation
Decomposition	Central figure	EA Gini points	ATT Gini points	EA channel shares
	ATT channel shares	Composition within P/A/B	Shapley vs one-factor	Not allocated
	Robustness			
Data	Sample selection	EUROMOD pricing	Numerical design	

The appendix map. Every entry is a link; every backup slide has a Back button to the main slide it supports and a button back to this map.

Backup: utility specification

$$v_i(j) = L_i(j) + \beta_c \log\left(\frac{C_i(j)}{\lambda_c}\right), \quad L_i(j) = \beta_\ell^g(\mathbf{x}_i) \mathcal{B}(\tilde{\ell}_i(j); \theta_\ell^g)$$

$$\tilde{\ell}_i(j) = \frac{80 - h}{10}, \quad \mathcal{B}(z; \theta) = \frac{z^\theta - 1}{\theta}$$

$$\beta_\ell^g(\mathbf{x}_i)$$

intercept, centred age and its square (units of 10 years)

$$k_i$$

children under 20, women only

couples

sum of the two spouses' leisure terms

The full systematic utility. Leisure is weekly time left from a maintained endowment, rescaled, and passed through a Box-Cox transform with a group-specific curvature. The leisure weight varies with centred age and, for women, with the number of children; that child term is a reduced-form time-constraint shifter, not pure taste. The consumption weight is estimated. The shock is extreme value with unit scale.

$$g_i(j) = g_i^E \cdot \left(g_i^H(h) \, g_i^{Occ}(k) \, g_i^W(w \mid k) \right)^{E_i(j)}$$

access

g_i^E : local unemployment exposure, region, urban or rural location, year

hours

g_i^H : elevated bands over a residual set

occupation

g_i^{Occ} : sex-specific group masses

wage offers

g_i^W : log-normal, truncated to [2, 590]

wage-offer location $\mu_i(k)$: education, potential experience and its square, occupation

The opportunity density factorises into access, switched on for employment, and three blocks for the employed part: hours, occupation and wage offers. The unemployment exposure enters access scaled by 10. The wage location depends on education, potential experience in units of 20 years and occupation, with a common dispersion. The hours density is a density elevation over intervals, not an atom at a particular hour.

Backup: sampled-alternative likelihood

$$V_{ij} = v_i(j) + \log g_i(j) - \log q_{ij}$$

$$\Pr(y \mid \mathcal{C}_i) = \frac{n_y \exp V_{iy}}{\sum_{s \in \mathcal{C}_i} \exp V_{is}}$$

q_{ij}
proposal density, fitted out of fold

$\mathcal{C}_i$
observed job and 100 draws: 101 rows

n_y
multiplicity of a repeated draw

The choice set is a continuum, so the likelihood is evaluated over sampled alternatives drawn from a proposal density and corrected by it. The proposal is a computational device: fitted out of fold so a household's own outcome never enters the proposal used to score it, and it enters no welfare measure. For couples the proposal draws the joint participation regime first.

Backup: identification

Choices alone do not separate a taste for leisure from a scarcity of jobs at that number of hours. Three restrictions do the work: a spike at a band of hours is read as availability rather than a kink in tastes; excluded shifters enter availability and not preferences; and wage offers are independent of hours given occupation. These are the restrictions of the latent-jobs literature, maintained here rather than tested. No causal effect of geography is claimed.

Functional form

- smooth preferences in hours;
- banded hours opportunities

Exclusion

local unemployment exposure and
geography shift access, not utility

Independence

wage offers independent of hours,
given occupation

maintained, not tested · not a causal design

Backup: preference parameters

The preference block. The consumption weight is common within each population. One single-adult coordinate, the female age-square term, sits at a box endpoint, so its standard error is not reported and it is excluded from the interior curvature. Levels are not comparable across the two separately estimated models.

	Single men	Single women	Couple men	Couple women
leisure weight, intercept	8.5199 (3.5480)	5.8683 (2.1346)	3.9814 (0.7466)	13.1029 (4.7760)
leisure weight, age	1.4820 (1.0236)	0.0678 (0.4792)	−0.0080 (0.0287)	−0.1228 (0.1260)
leisure weight, age squared	0.7783 (0.7732)	1.0000 (at bound)	0.0065 (0.0030)	0.0073 (0.0103)
leisure weight, children	—	0.1666 (0.4422)	—	−0.3055 (1.1472)
leisure curvature θ_ℓ	−1.6263 (0.3301)	−0.9274 (0.2161)	−0.9761 (0.1357)	−1.6863 (0.2458)
consumption weight β_c	2.0387 (0.2917)		2.1017 (0.2939)	

cluster-robust standard errors; the male children term in couples is structurally zero

Backup: employment-access parameters

	Singles	Couples
employment constant	−3.1735 (0.4053)	men −2.1233 (0.3195); women −2.8148 (0.3021)
local unemployment exposure	−1.4422 (0.2356)	−1.1924 (0.1557)
densely populated	−0.0288 (0.2209)	−0.1823 (0.1657)
intermediate density	0.0641 (0.2614)	−0.4174 (0.1861)
Region 2	−0.3780 (0.3292)	−0.1546 (0.2442)
Region 3	−0.1117 (0.3891)	0.0873 (0.2808)
Region 4	−0.8218 (0.3810)	0.0150 (0.3068)
Region 5	−0.5162 (0.3274)	−0.1455 (0.2555)
Region 6	−0.7222 (0.3528)	−0.2844 (0.2748)
Region 7	−0.5374 (0.3503)	−0.1396 (0.2651)
Region 8	−0.4512 (0.3401)	−0.1659 (0.2660)

references: the first region and thinly populated areas; couples share the shifters across spouses

The employment-access block. The employment index multiplies the working indicator. Local unemployment exposure lowers employment opportunity mass in both populations. Regional and urbanisation indicators shift access conditional on the exclusion restriction; they are not causal effects of geography.

Backup: hours and occupation opportunities

Hours bands				
	Singles	Couple men	Couple women	
part-time lower	0.0298	−1.1094	−0.5173	
part-time upper	0.7080	0.0835	1.1586	
narrow full-time	2.0656	2.2888	2.0284	
full-time upper	1.9387	2.3801	1.6825	
long hours	−0.0834	0.6977	−0.3109	
Occupation masses				
	Singles		Couples	
	men	women	men	women
Occupation 2	−1.3737	0.1135	−1.5097	0.2074
Occupation 3	−2.1096	−0.4060	−2.2532	−0.1804
Occupation 4	−0.3259	0.5632	0.1594	0.8159

estimates only; standard errors in the paper. References: the residual hours set and the craft and elementary occupation group

Hours opportunities are elevated densities over five bands relative to a residual set; hours coefficients are common across single men and women and spouse-specific for couples. Occupation masses are sex-specific, relative to the craft, agricultural, operator and elementary group. Standard errors for these blocks are in the paper’s tables.

Backup: wage-offer equation

	Singles	Couples
intercept	2.0138 (0.0561)	2.0480 (0.0335)
low education	0.0566 (0.0372)	−0.0433 (0.0217)
high education	0.1491 (0.0308)	0.1817 (0.0197)
potential experience	0.2408 (0.0862)	0.5643 (0.0546)
experience squared	−0.0282 (0.0390)	−0.1598 (0.0243)
Occupation 2	−0.0347 (0.0399)	−0.0740 (0.0231)
Occupation 3	0.0622 (0.0383)	0.0394 (0.0228)
Occupation 4	0.2785 (0.0368)	0.2146 (0.0223)
log-wage dispersion σ	0.3815 (0.0133)	0.3631 (0.0072)

medium education is the reference; experience in units of 20 years

The wage-offer equation. The offered log wage is normal with this location and a common dispersion, truncated to the wage support and renormalised. The singles and couples models are estimated separately, so the two experience profiles are not restricted to agree and their difference is not a test.

Backup: common versus household-specific opportunities

The two re-estimated common-opportunity benchmarks restrict the opportunity block and re-estimate everything else. Both are worse on the same criterion, and population fit worsens where the opportunity block does its work. A better criterion does not by itself establish that a mechanism has been identified.

Single adults	free coordinates	criterion	population fit
household-specific opportunities	41	6253.463	0.0140
common opportunity distribution	10	6403.974	0.0273
common shape; employment and hours in utility	16	6395.108	0.0264

same households, sampled alternatives and criterion; not converted into a formal test

Single adults	Couples
free coordinates 41, interior 40	free coordinates 47, interior 47
criterion 6253.463	criterion 10283.034
smallest Hessian eigenvalue 0.1669	smallest Hessian eigenvalue 0.0771
cluster-robust standard errors on the household	eigenvalues on the interior block

Curvature and inference. The exact Hessian on the interior block is positive definite for both models. Several terminal paths from different starts agree to numerical precision; that supports a stable local solution from the starts tested, not global uniqueness. One single-adult coordinate at a box endpoint is handled under the active-set convention.

Backup: extensive-margin fit

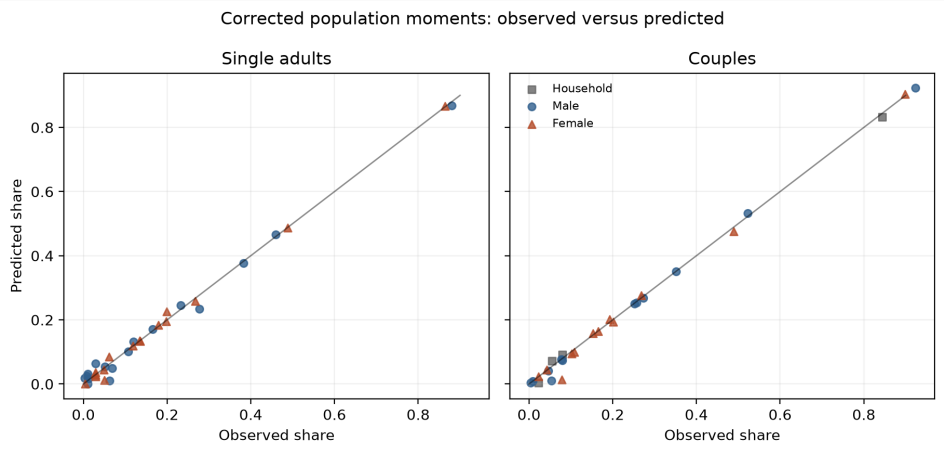
Extensive-margin accuracy is reportable for women in each sample and withheld for men, because numerical integration error is too large relative to sampling variation. The integration ratio is that comparison; the rule was fixed before the evaluation.

	weighted accuracy	model-simulated range	integration ratio
single women	85.7%	80.0–85.9%	0.096
coupled women	89.8%	88.1–91.2%	0.049
single men	withheld	—	0.361
coupled men	withheld	—	0.353

withheld where numerical integration error is too large relative to sampling variation

Backup: fit by margin

Corrected population fit: observed against predicted shares, margin by margin. Participation and occupation margins are reproduced closely in aggregate; the hours fit is uneven. Population predictions integrate the model over opportunities and taste shocks, so this is not an in-sample fitted choice.



Backup: the thirty-seven-hour point

single men

5.2

pp

single women

3.8

pp

coupled men

4.4

pp

coupled women

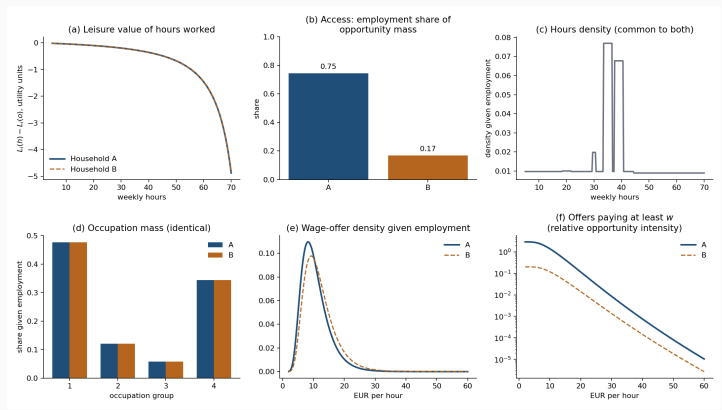
6.5

pp

observed minus predicted share at the observed thirty-seven-hour mass point

The common hours mismatch is underprediction of the observed concentration at exactly thirty-seven hours. It is a finding about that mass point, not about the model's neighbouring full-time range, and it is kept as its own result.

Backup: matched households

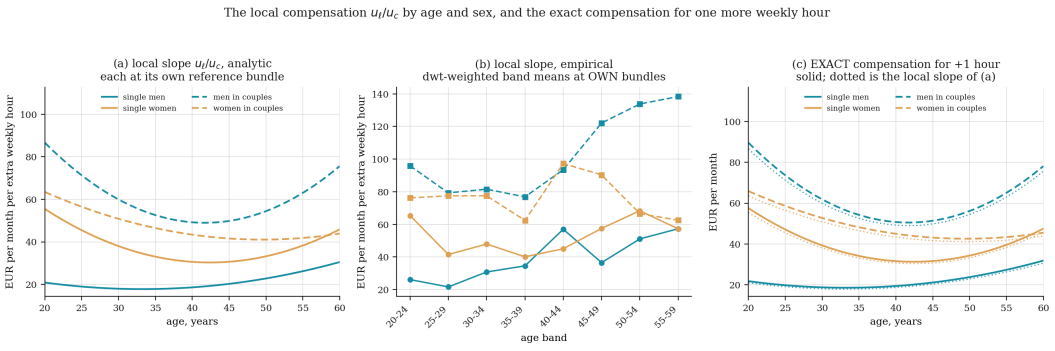


access: $A = 14.5 \times B$ wage-offer location: $B + 11.5 \text{ log points}$

What unequal opportunity means inside the model. Two employed single men in the same occupation group, hours band and wage quintile, with nearly identical estimated leisure profiles, selected by a stated rule: a teaching example, not causal and not representative. Employment is 0.75 of A's opportunity mass and 0.17 of B's. B's wage offers are better, yet A faces more offers paying at least any given wage over essentially all offer mass. Neither is better placed on every margin.

Backup: marginal rate of substitution

The marginal rate of substitution between leisure and consumption, by age and sex, in euros per month per additional recurring weekly hour of leisure. It is a compensation along an indifference curve, not a behavioural response to a wage change. The coefficients are coordinates; this is the economics.



Backup: attained-bundle derivation

$$L_i(o) + \beta_c \log\left(\frac{M_i^{\text{att}}}{\lambda_c}\right) = L_i(j_i^{\text{obs}}) + \beta_c \log\left(\frac{C_i^{\text{obs}}}{\lambda_c}\right)$$

$$\implies M_i^{\text{att}} = C_i^{\text{obs}} \exp\left\{\frac{L_i(j_i^{\text{obs}}) - L_i(o)}{\beta_c}\right\}$$

Reference job

the preferred job when every job pays the same:
here non-employment for every household

Staying-home equivalent

a verified property of this specification, not a
general theorem

The attained-bundle money metric solves an indifference condition between the observed job and consumption and the reference state at the unknown consumption. The normaliser cancels. The general construction uses the job the household would most prefer at equal pay; on the current domain non-employment maximises the non-consumption index for every household, so the measure is the staying-home equivalent. With a different leisure specification the two could separate.

Backup: ex-ante derivation

$$J_i = \int_{\Omega_i} e^{L_i(j)} \left(\frac{C_i(j)}{\lambda_c} \right)^{\beta_c} \widehat{g}_i(j) d\nu(j)$$

$$J_i^{\text{ref}}(m) = \left(\frac{m}{\lambda_c} \right)^{\beta_c} H_i, \quad H_i = \int_{\Omega_i} e^{L_i(j)} \widehat{g}_i(j) d\nu(j)$$

$$J_i^{\text{ref}}(M_i^{\text{EA}}) = J_i \implies M_i^{\text{EA}} = \lambda_c \exp \left\{ \frac{\log J_i - \log H_i}{\beta_c} \right\}$$

a common rescaling of $\widehat{g}_i$ cancels $\cdot \log J_i$: expected utility of the best reachable job, up to a constant

The reference prospect keeps the household's jobs, availability and preferences and pays the same consumption at every job; its value is the power of that consumption times H. Setting it equal to the actual prospect and solving gives the ex-ante money metric. Under the extreme-value shocks, $\log J$ is the expected utility of the best reachable job up to a constant common to J and H, which is why taste-shock variety counts as welfare-relevant: a normative position.

Backup: ex-ante numerical validation

Identities

constant-consumption identity

one-state identity

common intensity-rescaling invariance

flat-reference inversion

exact Shapley adding-up

Numerics

exact reference integral on the priced se

effective sample size and weight concentration

two-seed stability of welfare, Gini and Shapley

finite positive welfare for every household

independent reimplementations

all passed for both populations · a correct computation, not causal identification or parameter uncertainty

The ex-ante calculation passed every declared check for both populations, including an independent reimplementa-tion, with the integration design fixed before inspecting any welfare level or Shapley value. That certifies the computation given the model and operators. It does not establish causal identification, it does not quantify statistical uncertainty in the estimated parameters, and it does not make either welfare perspective normatively correct. Channel orderings are unchanged across the reported reference-domain sensitivities.

Backup: equivalisation

$$M_i^{p,eq} = \frac{M_i^p}{e_i}, \quad e_i : \text{ modified-OECD scale}$$

Couples, ex ante: **A + B**

21.3% $\longrightarrow$ 7.9%

raw $\rightarrow$ equivalised; % of baseline Gini

Couples, ex ante: **preferences**

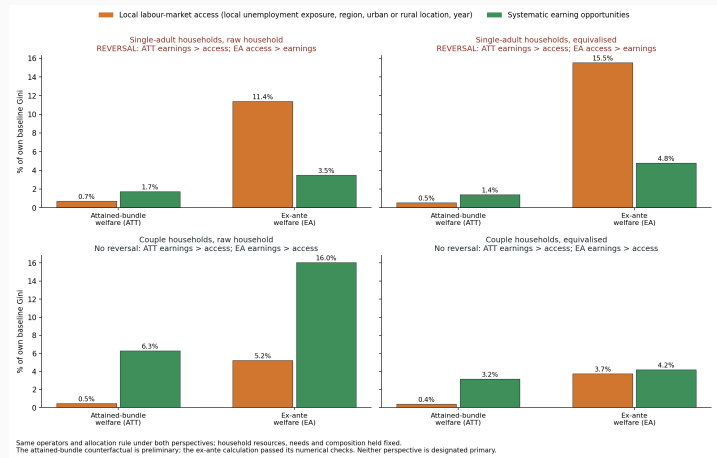
0.00291 $\longrightarrow$ -0.00212

Gini points, raw $\rightarrow$ equivalised

a normative convention, reported both ways; populations never combined or compared

Equivalisation is a normative convention and it is consequential for couples under the ex-ante perspective: it moves access plus earnings as a share of baseline inequality and turns the preference contribution negative. A negative Shapley term means that, averaged over orders, equalising preferences raises inequality; it does not show that preference heterogeneity is equalising in any causal or welfare sense. That is why both scales are always reported.

Backup: central figure



The published central figure: access and earning-opportunity contributions, each as a percentage of its own perspective's baseline Gini, under attained-bundle and ex-ante welfare, raw and equivalised. For single adults earning opportunities are larger under the attained bundle and access is larger under the prospect; couples show no reversal. Resources, needs and composition are held fixed.

Backup: ex-ante decomposition in Gini points

The full ex-ante table. Contributions are signed Gini points; access plus earnings is reported relative to each baseline Gini. The calculation passed its numerical checks, including an independent implementation. For couples the preference contribution changes sign with equivalisation.

	scale	baseline Gini	P	A	B	A + B, % of baseline	ordering
Single adults	raw	0.1595	0.00157	0.01812	0.00555	14.8	access > earnings
	equivalised	0.1406	0.00020	0.02183	0.00673	20.3	access > earnings
Couples	raw	0.1277	0.00291	0.00667	0.02049	21.3	earnings > access
	equivalised	0.1200	−0.00212	0.00449	0.00504	7.9	earnings > access

exact Shapley contributions in Gini points; household-weighted

Backup: attained-bundle decomposition in Gini points

The attained-bundle table. Earning opportunities are larger than local access for both household types and both scales. The preference contribution changes sign with equivalisation in both samples, so no directional claim is made about it. The lower part of the hours range is sparsely represented in the integration sample, which is why this exercise stays preliminary.

	scale	baseline Gini	P	A	B	A + B, % of baseline	ordering
Single adults	raw	0.2337	0.0012	0.0016	0.0040	2.4	earnings > access
	equivalised	0.2451	−0.0003	0.0013	0.0034	1.9	earnings > access
Couples	raw	0.2035	0.0064	0.0009	0.0128	6.7	earnings > access
	equivalised	0.1971	−0.0019	0.0007	0.0062	3.5	earnings > access

exact Shapley contributions in Gini points; preliminary; household-weighted

Backup: ex-ante channels, % of baseline

Each ex-ante Shapley contribution, in Gini points and as a percentage of its own baseline Gini, raw and equivalised. The negative couples preference term is kept. P, A and B do not exhaust baseline inequality: household resources, needs and composition and other dimensions are held fixed or left unallocated.

		P				A		B	
	scale	I_0	Gini pts	% of I_0	Gini pts	% of I_0	Gini pts	% of I_0	
Single adults	raw	0.1595	0.00157	0.99	0.01812	11.37	0.00555	3.48	
	equivalised	0.1406	0.00020	0.14	0.02183	15.52	0.00673	4.79	
Couples	raw	0.1277	0.00291	2.28	0.00667	5.22	0.02049	16.04	
	equivalised	0.1200	−0.00212	−1.76	0.00449	3.75	0.00504	4.20	

each contribution divided by its own baseline Gini I_0 ; signed; the rows are not forced to sum to one hundred, because the remainder is not allocated by this exercise

Backup: attained-bundle channels, % of baseline

		P			A		B	
	scale	I_0	Gini pts	% of I_0	Gini pts	% of I_0	Gini pts	% of I_0
Single adults	raw	0.2337	0.0012	0.53	0.0016	0.69	0.0040	1.73
	equivalised	0.2451	−0.0003	−0.13	0.0013	0.54	0.0034	1.40
Couples	raw	0.2035	0.0064	3.16	0.0009	0.45	0.0128	6.28
	equivalised	0.1971	−0.0019	−0.98	0.0007	0.37	0.0062	3.15

each contribution divided by its own baseline Gini I_0 ; signed; preliminary

The same layout for the attained-bundle perspective. Earning opportunities exceed access in every row; the preference contribution changes sign with equivalisation. These percentages are shares of each baseline Gini and are not forced to sum to one hundred.

Backup: composition of the explained component

$$s_k = \frac{\phi_k}{\phi_P + \phi_A + \phi_B}, \quad k \in \{P, A, B\}$$

ex-ante prospect, equivalised	Single adults	Couples
Preferences s_P	0.7%	−28.5%
Local access s_A	75.9%	60.6%
Earning opportunities s_B	23.4%	67.9%
Total	100.0%	100.0%

These are shares of the currently explained P/A/B change, not shares of total inequality. Negative components are possible under Shapley attribution.

These are shares of the currently explained P/A/B change, not shares of total inequality. Negative components are possible under Shapley attribution.

Within that currently explained component, most of the single-adult opportunity effect comes through access rather than earning opportunities. For couples the negative preference share means the access and earning shares together exceed the whole explained change. Only the equivalised scale is shown: on the raw scale the one-decimal couples shares would not close exactly.

Backup: Shapley versus one-factor equalisation

one-factor: $v^P(\{k\}) = I^P(\emptyset) - I^P(\{k\})$

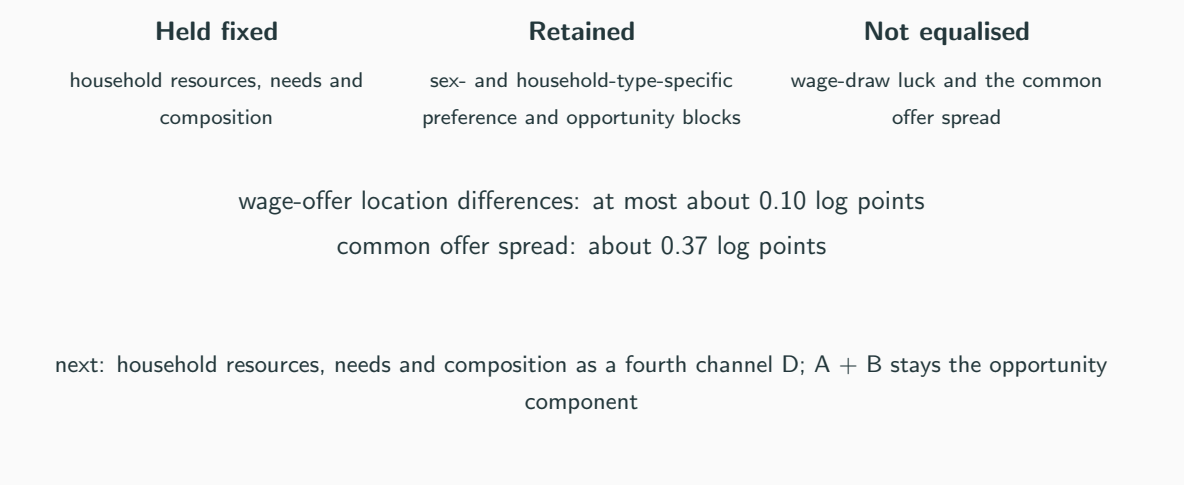
Shapley: $\phi_k^P = \text{average over all orders of } v^P(S \cup \{k\}) - v^P(S)$

$$\phi_P^P + \phi_A^P + \phi_B^P = I^P(\emptyset) - I^P(\{P, A, B\})$$

one-factor effects need not add up · Shapley contributions do, exactly

A one-factor effect equalises one channel from the actual situation. Because the channels interact, those effects need not add up, and the answer would depend on the order chosen. The Shapley contribution averages each channel's marginal effect over every order and adds up exactly to the total change. This is the standard rule from the distributional decomposition literature; no new principle is claimed. Coalition-by-coalition Gini tables are not reproduced in these slides.

Backup: what is not allocated



The not-allocated remainder is not a channel. It holds household resources, needs and composition fixed, it keeps the sex- and household-type-specific blocks, and it excludes wage-draw luck and the common offer spread. The location differences that the earning operator equalises are small next to that common spread, so the operator removes only a small part of earnings dispersion by construction; that is a definitional boundary, not a measurement error. Adding a fourth channel for resources, needs and composition is the planned extension; with it, access plus earnings would still be the opportunity component, and the broader sum would not be an opportunity share.

Backup: robustness of the attained-bundle decomposition

Second simulation seed

every coalition Gini and contribution
reproduces closely, including the preference
sign change

Observed job in the simulation

attained under every coalition by 2.9% of single
adults and 3.9% of couples
excluding it moves the total change by at most
9.3%, with no sign change

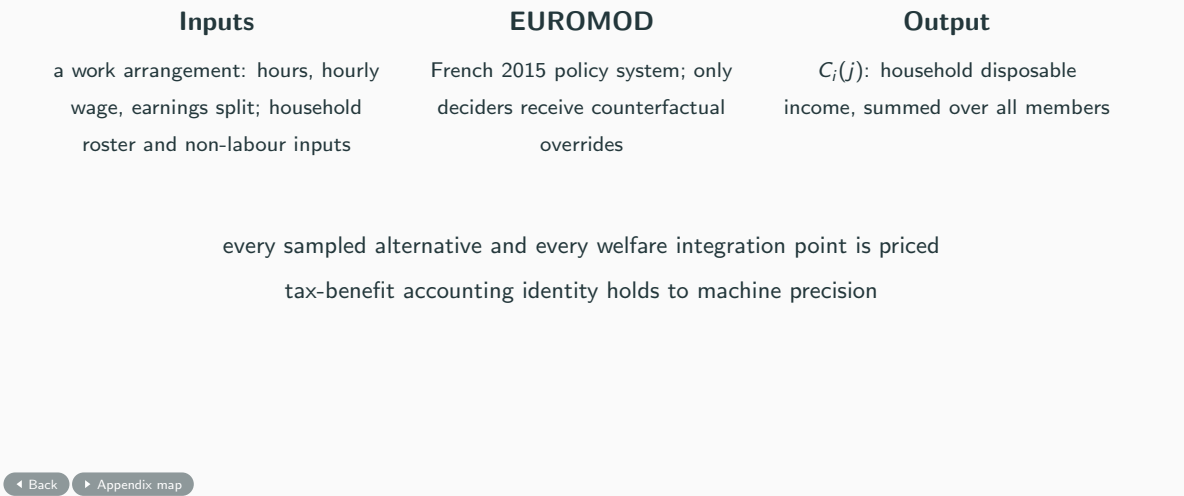
simulation checks, not confidence intervals

Robustness of the attained-bundle calculation. An independent second seed reproduces every coalition Gini and Shapley contribution closely. The simulation always includes each household’s own observed job; excluding it moves the total change by at most the amount shown, in the couples equivalised case, with no sign change. These checks validate the calculation for the stated game, not its maintained assumptions, and they are not confidence intervals.

Backup: sample selection

Unweighted households remaining after each screen. The largest exclusion is structural: multi-generational households, flat-shares and same-sex couples are outside the estimated population, and no result extends to them. Students and pension recipients are removed because the model does not define an offer set for them; the remaining screens are support restrictions.

screen	single-adult	couple
the raw France 2016 input file	4,038	5,965
one- or two-adult households (composition screen)	4,038	5,965
every adult aged 20 to 60	2,131	3,662
no adult still in education	2,036	3,521
no old-age, disability or survivor pension receipt	1,755	3,218
labour status in scope	1,598	2,412
no other employable or earning member	1,564	2,323
hours and wage inside the modelled support	1,555	2,275
hours outside [5,70] or unsupported military occupation (ISCO 0) on an employed decider	1,543	2,223
observed chosen alternative priced at non-positive disposable consumption	1,540	2,223
estimation sample	1,540	2,223



For a given work arrangement the tax-benefit model receives the implied gross labour inputs together with the household's non-labour inputs and roster, and returns disposable income. Only deciders receive counterfactual overrides; other members keep their baseline values. The accounting identity linking original income, benefits, taxes and contributions to disposable income holds to machine precision at the person and household-alternative level.

Estimation	Predictive fit	Supports
101	2,048	[5, 70]
rows per household: the observed job and 100 sampled alternatives	common integration points per household	weekly hours; wages in [2, 590] per hour
sampling devices are computational: they are not offers and enter no welfare measure		

The numerical design. Estimation uses the observed job plus sampled alternatives per household. The predictive fit integrates over a separate, larger common set of priced integration points per household. The supports are the intervals on which the opportunity densities are defined. The ex-ante welfare integrals use their own certified design, chosen before any welfare result was inspected.